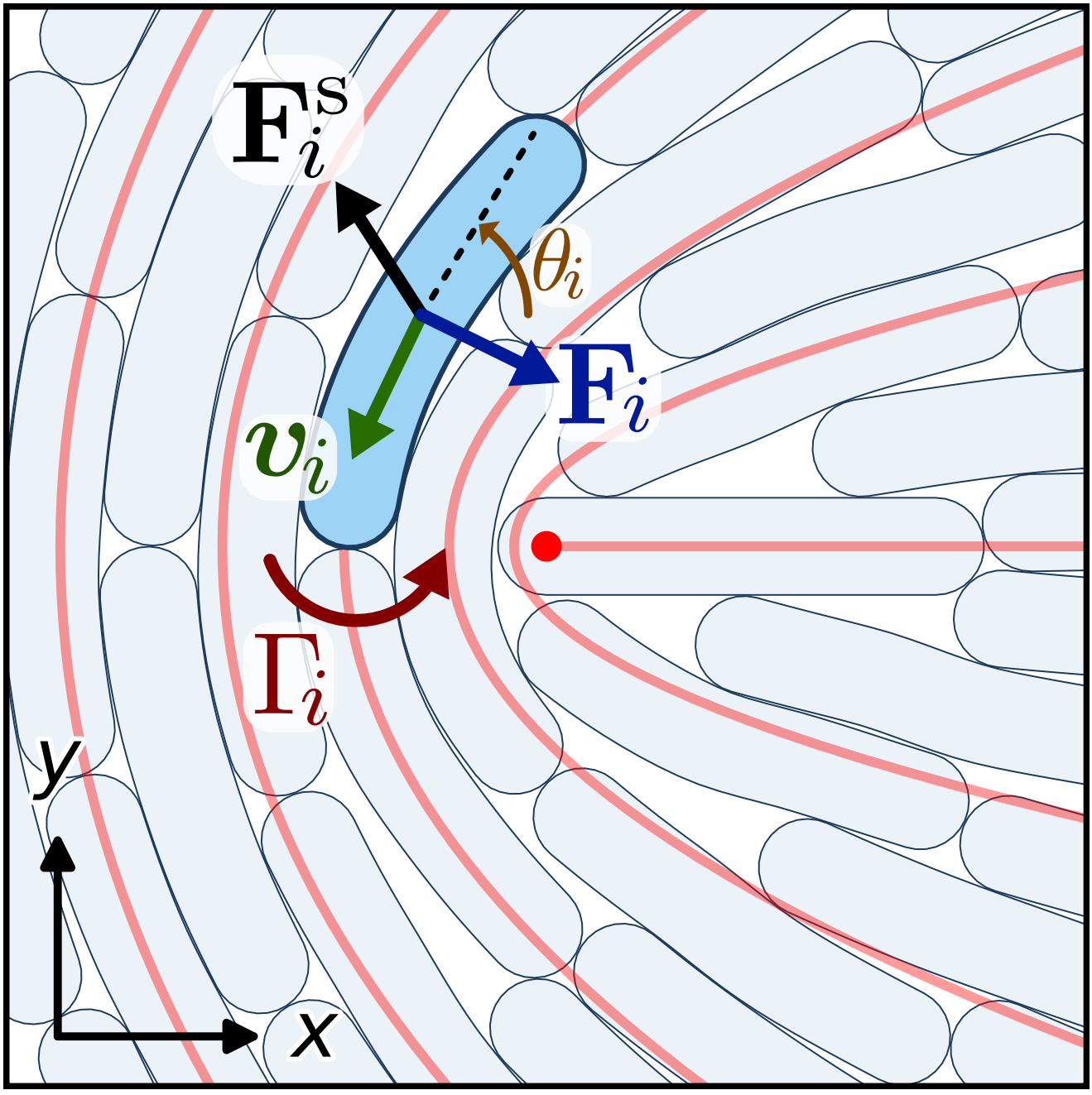
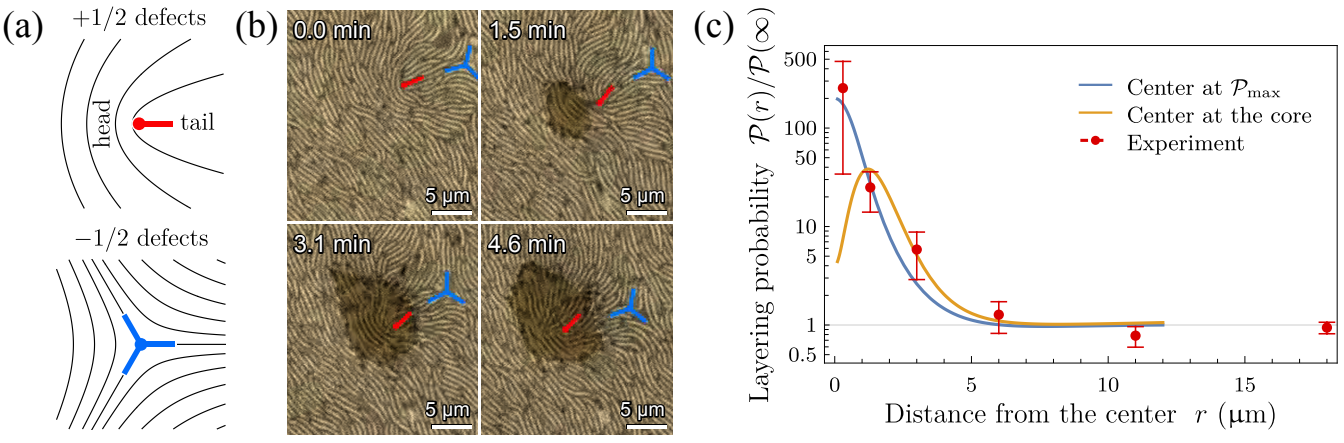
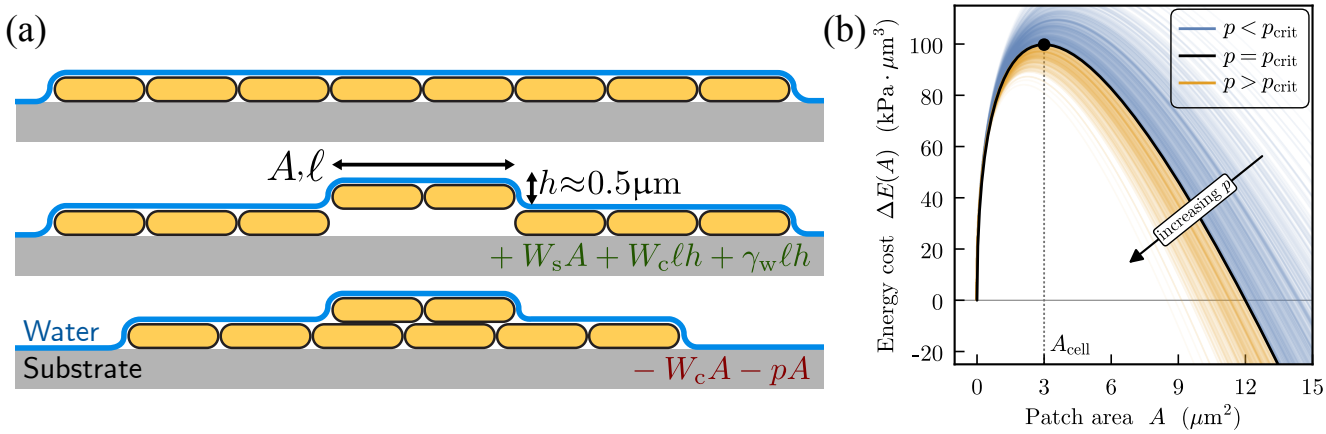

$\mathbf{F}_i^{\mathrm{S}}$
θ_i
$\mathbf{F}_i$
$\boldsymbol{v}_i$
Γ_i
y
x


(a)
+1/2 defects
head
tail
−1/2 defects
(b)
0.0 min
1.5 min
3.1 min
4.6 min
5 μm
(c)
Center at $\mathcal{P}_{\max}$
Center at the core
Experiment
Layering probability $\mathcal{P}(r)/\mathcal{P}(\infty)$
Distance from the center r (μm)
500
100
50
10
5
1
0.5
0
5
10
15


(a)
A,ℓ
h≈0.5μm
+ W_s A + W_c ℓh + γ_w ℓh
Water
Substrate
− W_c A − pA
(b)
p < p_crit
p = p_crit
p > p_crit
increasing p
Energy cost ΔE(A) (kPa · μm³)
Patch area A (μm²)
A_cell
100
80
60
40
20
0
-20
0
3
6
9
12
15